

Oxygen transport practice test

The O₂-hemoglobin dissociation curve, the Bohr effect, fetal hemoglobin and carbon monoxide.

Name _____ Date _____

Start time _____ End time _____

TIME LIMIT	50 minutes. Stop when time is up and mark where you were.
TOTAL POINTS	40 points · A 16 · B 6 · C 7 · D 11
ALLOWED	Pencil, eraser, scientific calculator. No notes, phone, or textbook.
COVERS	O1 Read the curve O2 Predict shifts and explain the Bohr effect O3 Apply affinity to fetal Hb, stored blood and CO O4 Calculate saturation and O ₂ content O5 Direction of O ₂ diffusion

Instructions

1. Take this test closed-book, timed, in one sitting.
2. Multiple choice: fill in one bubble. Choose the single best answer.
3. Written answers: use the space given. It is sized for a complete answer.
4. If you check notes or the word bank for an item, finish it, then circle its number. That item is scored "with notes," not unaided.
5. Assume standard conditions (pH 7.40, PCO₂ 40 mmHg, 37 °C) unless an item says otherwise.
6. The answer key starts on a new page marked ANSWER KEY. Remove it before you start.

After you finish: score with the key, then fill in the scoring sheet and the error-analysis page. Or photograph every page, flat and in good light, and upload the photos to be graded.

Section A · Multiple choice

8 items · 2 pts each

- 1 A blood sample's P50 rises from 27 to 33 mmHg. Its hemoglobin concentration does not change. Which statement about the sample is correct? [2 pts]

 - Ⓐ Its O₂ affinity increased, so it releases less O₂ in the tissues.
 - Ⓑ Its O₂ affinity decreased, so it releases more O₂ at a tissue PO₂ of 40 mmHg.
 - Ⓒ Its maximum O₂-carrying capacity fell, because fewer heme sites are available.
 - Ⓓ Its saturation at a PO₂ of 100 mmHg falls to about half of normal.
- 2 Why is the O₂-hemoglobin dissociation curve sigmoid rather than hyperbolic? [2 pts]

 - Ⓐ Each of the four hemes has a different fixed affinity, and they fill one at a time from highest to lowest.
 - Ⓑ O₂ must first dissolve in plasma, and plasma becomes saturated with O₂ at about 60 mmHg.
 - Ⓒ 2,3-BPG binds after the first O₂ and speeds the loading of the rest.
 - Ⓓ O₂ binding at one heme raises the O₂ affinity of the remaining hemes in the same tetramer.
- 3 Compared with the same muscle at rest, which combination of local changes in a sprinting quadriceps shifts the O₂-Hb curve of its capillary blood to the right? [2 pts]

 - Ⓐ PCO₂ down, pH up, temperature up
 - Ⓑ PCO₂ up, pH up, temperature down
 - Ⓒ PCO₂ up, pH down, temperature up
 - Ⓓ PCO₂ up, pH down, temperature down
- 4 Fetal blood leaving the placenta can carry more O₂ per deciliter than the maternal blood leaving the same exchange site, yet O₂ still moves from maternal to fetal blood. Which quantity sets the direction of net O₂ diffusion? [2 pts]

 - Ⓐ Partial pressure of O₂ (PO₂)
 - Ⓑ O₂ content (mL O₂ per dL blood)
 - Ⓒ Percent hemoglobin saturation
 - Ⓓ Hemoglobin concentration

- 5 In the placenta, maternal HbA and fetal HbF are exposed to the same PO_2 of about 30 mmHg. Compared with HbA, how saturated is HbF at that PO_2 , and why? [2 pts]
- Ⓐ Less saturated, because HbF has lower O_2 affinity so it can release O_2 to fetal tissues.
 - Ⓑ More saturated, because HbF binds 2,3-BPG weakly and so has higher O_2 affinity.
 - Ⓒ Equally saturated; the fetus gains O_2 only because it has more hemoglobin.
 - Ⓓ More saturated, because HbF binds 2,3-BPG more strongly than HbA does.
- 6 A climber flies to 4,300 m and within minutes is hyperventilating. Over the following days, red-cell 2,3-BPG rises. What happens to the position of the O_2 -Hb curve in the first 30 minutes? [2 pts]
- Ⓐ It shifts left, because lower PCO_2 and higher pH raise hemoglobin's O_2 affinity.
 - Ⓑ It shifts right, because altitude raises 2,3-BPG.
 - Ⓒ It shifts right, because the low inspired PO_2 lowers hemoglobin's O_2 affinity.
 - Ⓓ It does not shift, because changes in PCO_2 and pH alter breathing, not hemoglobin.
- 7 A patient pulled from a house fire has 30% carboxyhemoglobin, hemoglobin 15 g/dL and arterial PO_2 100 mmHg. Which statement best describes this patient's arterial blood compared with normal? [2 pts]
- Ⓐ PaO_2 is low, because CO displaces O_2 from the plasma.
 - Ⓑ O_2 content is low, but the remaining Hb releases O_2 more easily (right shift), which partly compensates.
 - Ⓒ PaO_2 is normal, O_2 content is low, and the remaining Hb holds its O_2 more tightly (left shift).
 - Ⓓ O_2 content is normal, because PaO_2 is normal.
- 8 A trauma patient receives packed red cells stored for about three weeks; their 2,3-BPG is nearly gone. In the first hours after transfusion, at the same tissue PO_2 , how do the transfused cells compare with the patient's own red cells? [2 pts]
- Ⓐ They release more O_2 , because low 2,3-BPG lowers hemoglobin's O_2 affinity.
 - Ⓑ They release the same O_2 , because the patient's pH and temperature set the curve.
 - Ⓒ They release less O_2 , because stored cells carry less hemoglobin per cell.
 - Ⓓ They release less O_2 , because low 2,3-BPG leaves their hemoglobin with higher O_2 affinity.

Section B · Short answer

2 items · 3 pts each

- 9 Name the effect by which blood acidity and CO_2 in metabolically active tissue change O_2 delivery, and explain in 2–3 sentences how it works at the hemoglobin molecule and what it means for that tissue. [3 pts]

- 10 Person X has anemia with hemoglobin 10.5 g/dL. Person Y has hemoglobin 15 g/dL with 30% carboxyhemoglobin. Both have reduced arterial O_2 content. Explain why Y's tissues are worse off than X's would be at a similar content. [3 pts]

- 11** Use the Hill equation, $S = P^n / (P^n + P50^n)$, with $n = 2.7$ and $P50 = 26.6$ mmHg. [4 pts]

(a) Calculate S at an arterial PO_2 of 100 mmHg. [1]
(b) Calculate S at a mixed venous PO_2 of 40 mmHg. [1]
(c) With hemoglobin 15 g/dL and 1.34 mL O_2 per g Hb, and ignoring dissolved O_2 , how much O_2 does each deciliter of blood unload between 100 and 40 mmHg? [2]

WORK

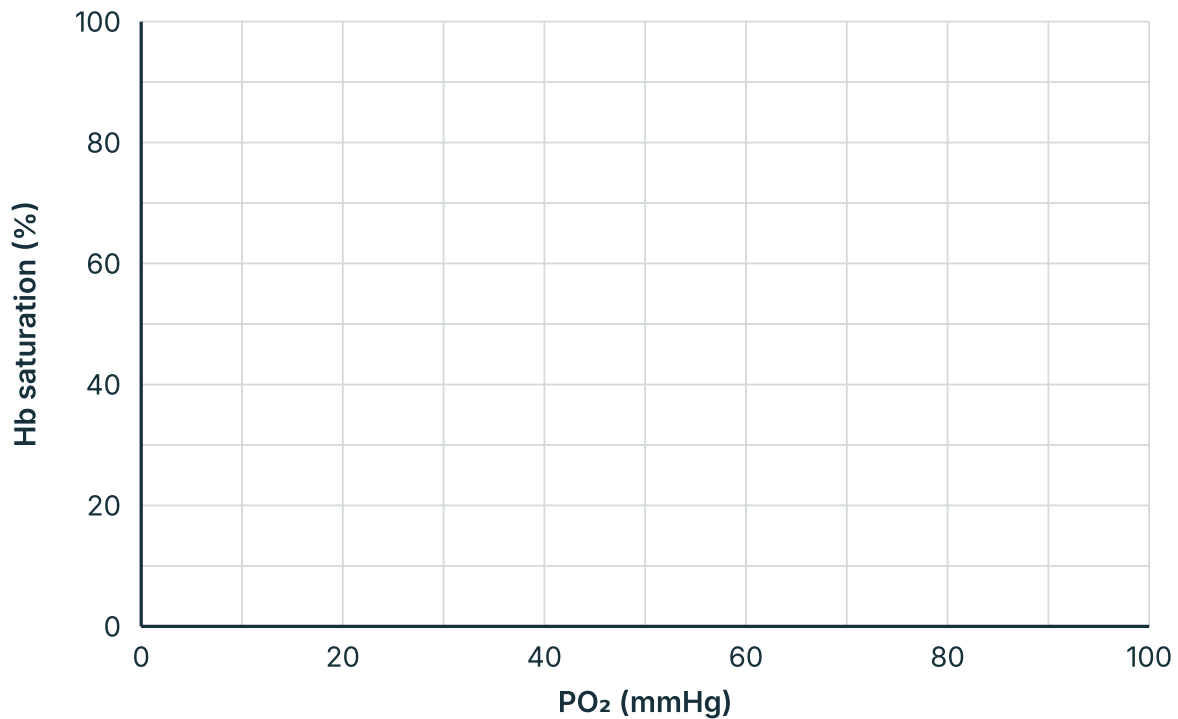
(a) _____ % (b) _____ % (c) _____ mL O_2 /dL

- 12** The house-fire patient from item 7 has hemoglobin 15 g/dL and 30% carboxyhemoglobin. Assume the hemoglobin not bound to CO is fully saturated with O_2 . Ignoring dissolved O_2 , calculate the arterial O_2 content, and compare it with the same person's content with no CO (also fully saturated). [3 pts]

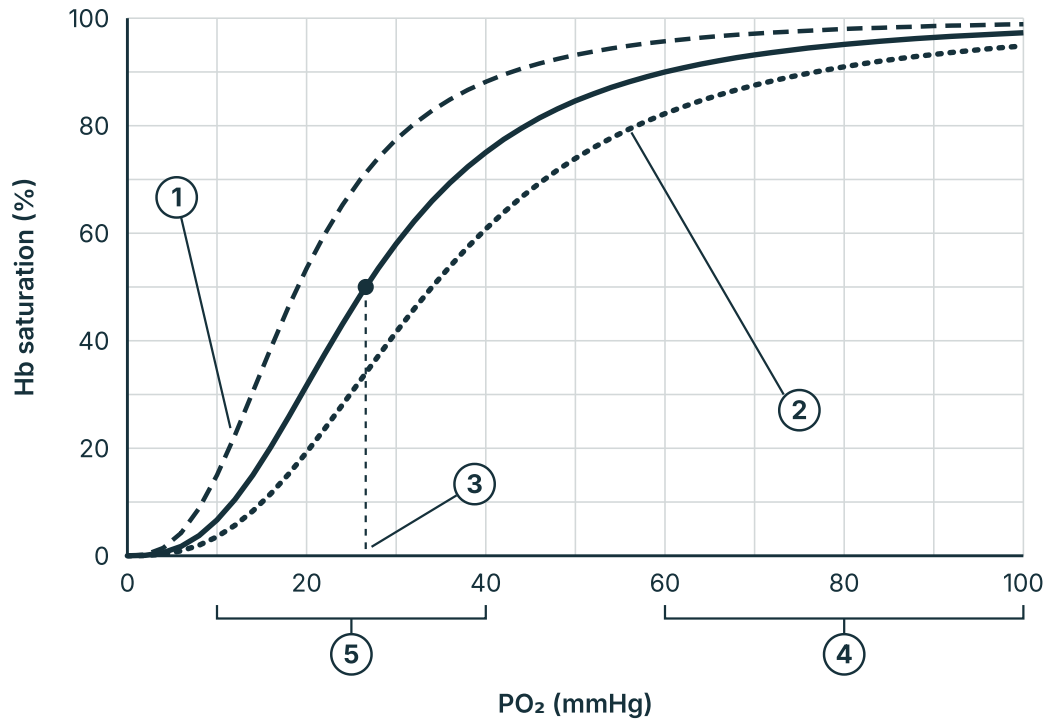
WORK

With CO _____ mL/dL Without CO _____ mL/dL Change _____ %

- 13** (a) With a solid line, sketch the normal adult O_2 -Hb curve. Mark and label P50, the arterial point (PO_2 100 mmHg) and the mixed venous point (PO_2 40 mmHg), each with its approximate saturation. [6 pts]
- (b) With a dashed line, sketch the curve for capillary blood in an exercising muscle. Draw an arrow at 40 mmHg showing the extra O_2 it unloads. [2]



- 14 The solid curve is normal adult blood at standard conditions. For each number, write [5 pts]
the name of the feature. For curves 1 and 2, also give one condition that produces that
curve (name and condition both needed for the point). For 3, give the approximate value.
For 4 and 5, say what that part of the solid curve means for O₂ loading or unloading.



1	
2	
3	
4	
5	

WORD BANK (cover it on your first try; using it makes the item "with notes"): left shift · right shift · P₅₀ · plateau (loading zone) · steep part (unloading zone) · Hill coefficient · venous point

End of test. Record your end time. Do not turn the page until you are done.

ANSWER KEY

Answer key and explanations

Score written items with the rubric, not by keywords. Values from the Hill model ($n = 2.7$, $P50 = 26.6$ mmHg) were computed, not read off a graph.

Blueprint

OBJECTIVE	ITEMS	POINTS	SHARE
O1 · Read the curve: shape, P50, plateau and steep part, arterial and venous points	1, 2, 13a, 14	13	33%
O2 · Predict shifts from PCO_2 , pH, temperature, 2,3-BPG; explain the Bohr effect	3, 6, 9, 13b	9	22%
O3 · Apply affinity to fetal Hb, stored blood and CO	5, 7, 8, 10	9	22%
O4 · Calculate saturation, O_2 content and unloading	11, 12	7	18%
O5 · Direction of O_2 diffusion (PO_2 gradient)	4	2	5%

Transfer items (new surface, same principle): 6, 8, 10, 12. In a course test, weights follow the study guide and each Source cites its line or slide.

Sources

1. **S1** Severinghaus JW. Simple, accurate equations for human blood O_2 dissociation computations. *J Appl Physiol* 1979;46:599–602. Standard curve, P50 26.6 mmHg.
2. **S2** A medical physiology text, O_2 transport chapter (e.g., Boron & Boulpaep, *Medical Physiology*; Hall, *Guyton and Hall Textbook of Medical Physiology*).
3. **S3** Roughton FJW, Darling RC. The effect of carbon monoxide on the oxyhemoglobin dissociation curve. *Am J Physiol* 1944;141:17–31.
4. **S4** Benesch R, Benesch RE. The effect of organic phosphates from the human erythrocyte on the allosteric properties of hemoglobin. *Biochem Biophys Res Commun* 1967;26:162–167.

Section A

- 1 **B.** P50 is the PO_2 that half-saturates Hb. Needing more PO_2 to get there means lower affinity (right shift). In the Hill model, saturation at 40 mmHg falls from about 74% to 63%, so more O_2 is released.
 - **A** (P50 read backwards): a higher number feels like "more binding." It means the opposite.
 - **C** (affinity confused with capacity): capacity depends on Hb amount, which did not change.
 - **D** (plateau ignored): at 100 mmHg the curve is flat; S drops only from about 97% to 95%.

O1 · S1, S2

2 D. Cooperativity: binding shifts the tetramer from the T (tense) toward the R (relaxed) state, raising the affinity of the remaining sites. The Hill coefficient of about 2.7 (greater than 1) measures this.

- **A** (independent sites): sites filling from highest to lowest affinity would flatten the curve, not make it sigmoid.
- **B** (plasma limits binding): dissolved O_2 rises linearly with PO_2 (0.003 mL/dL per mmHg); plasma does not saturate.
- **C** (2,3-BPG role reversed): 2,3-BPG binds deoxy-Hb and lowers affinity.

O1 · S2

3 C. Working muscle makes CO_2 , acid and heat. All three lower affinity and shift the curve right, so more O_2 unloads where it is needed.

- **A** (CO_2 thought to be consumed): muscle produces CO_2 ; falling PCO_2 shifts left.
- **B** (pH scale read backwards): more H^+ means lower pH; and cooling shifts left.
- **D** (temperature effect reversed): heat lowers affinity; cooling raises it.

O2 · S2

4 A. Gases diffuse down their partial-pressure gradient. Only dissolved O_2 diffuses, and its amount is set by PO_2 . Hb-bound O_2 adds to content but does not drive diffusion.

- **B** (content gradient): the stem's own case refutes it: fetal content can be higher while O_2 still flows in.
- **C** (saturation as driving force): saturation reflects affinity as well as PO_2 .
- **D** (Hb pulls O_2): Hb raises carrying capacity; it does not set direction.

O5 · S2

5 B. HbF's γ chains bind 2,3-BPG weakly, so HbF escapes that right shift (P_{50} about 19 vs 26.6 mmHg). In the model at 30 mmHg: HbF about 77%, HbA about 58%. High-affinity HbF keeps fetal plasma PO_2 low, so the maternal-to-fetal gradient persists.

- **A** (fetal Hb has lower affinity): reasoning from "the fetus must release O_2 " gets the direction backwards.
- **C** (only Hb amount matters): fetal Hb concentration is higher, but affinity also differs.
- **D** (2,3-BPG effect reversed): stronger 2,3-BPG binding would lower affinity.

O3 · S2, S4

- 6 **A.** Within minutes, hyperventilation causes respiratory alkalosis, so lower PCO_2 and higher pH shift the curve left. The 2,3-BPG rise (a right shift) builds over hours to days.
- **B** (time course mixed up): true for days later, not the first 30 minutes.
 - **C** (PO_2 treated as a shifter): a lower PO_2 moves the point down the curve; it does not shift the curve.
 - **D** (blood chemistry thought not to touch hemoglobin): blowing off CO_2 lowers H^+ ($\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{CO}_3 \rightleftharpoons \text{H}^+ + \text{HCO}_3^-$), and less CO_2 and H^+ bound to Hb raises its O_2 affinity (Bohr effect in reverse).

O2 · transfer · S2

- 7 **C.** CO binds heme with roughly 200–250 times O_2 's affinity, cutting O_2 content. PaO_2 measures dissolved O_2 , so it stays normal. CO also raises the O_2 affinity of the remaining sites, a left shift that impairs unloading.
- **A** (PaO_2 confused with bound O_2): CO competes for heme, not for dissolved O_2 .
 - **B** (CO shifts right): the shift is left, which makes things worse.
 - **D** ($\text{PaO}_2 = \text{content}$): content depends on functional Hb and saturation.

O3 · S3

- 8 **D.** 2,3-BPG lives inside each red cell. Without it, transfused cells are left-shifted and hold their O_2 until they rebuild 2,3-BPG over the following days.
- **A** (direction of the 2,3-BPG effect reversed): 2,3-BPG lowers affinity, so losing it *raises* affinity.
 - **B** (only the environment sets the curve): intracellular 2,3-BPG travels with the cell.
 - **C** (content confused with affinity): the cells' Hb per cell is not why they release less.

O3 · transfer · S2, S4

Section B

- 9 **Model answer.** The Bohr effect. H^+ binds hemoglobin (protonating histidine residues) and CO_2 binds its amino ends as carbamino groups; both stabilize the deoxy (T) state and lower O_2 affinity. The curve shifts right, so at the same tissue PO_2 less O_2 stays bound and more is unloaded where metabolism is highest.

1. 1 pt: names the Bohr effect, or says right shift / lower affinity.
2. 1 pt: mechanism: H^+ and/or CO_2 bind Hb and favor the deoxy (T) state.
3. 1 pt: consequence: lower saturation at the same tissue PO_2 , so more O_2 unloaded in that muscle.

O2 · S2 (Bohr, Hasselbalch & Krogh 1904)

10 Model answer. Both have about 10.5 g/dL of Hb able to carry O₂ (15 × 0.70), so content is similar. But in Y, CO raises the O₂ affinity of the remaining sites (left shift). Y's Hb gives up less O₂ at a given tissue PO₂, so tissue PO₂ must fall lower before the O₂ is released.

1. 1 pt: functional Hb, and so content, is similar.
2. 1 pt: CO left-shifts the curve of the remaining Hb (raises affinity).
3. 1 pt: so less O₂ unloads at tissue PO₂ (or tissue PO₂ must fall lower). Do not require CO's effect on cytochrome oxidase; it is a true extra.

O3 · transfer · S3

Section C

11 (a) 97% · (100/26.6)^{2.7} = 3.759^{2.7} ≈ 35.7; S = 35.7 / 36.7 ≈ 0.973.

(b) 75% · (40/26.6)^{2.7} = 1.504^{2.7} ≈ 3.01; S = 3.01 / 4.01 ≈ 0.751.

(c) ≈ 4.5 mL O₂/dL · capacity 1.34 × 15 = 20.1 mL/dL; 20.1 × (0.973 – 0.751) ≈ 4.47 mL/dL.

1. (a) 1 pt: 97% (accept 97–98%). (b) 1 pt: 75% (accept 74–76%).
2. (c) 1 pt: setup, capacity × difference in saturation. 1 pt: 4.4–4.6 mL O₂/dL with units.

O4 · S1 (P50), S2 (1.34 mL/g)

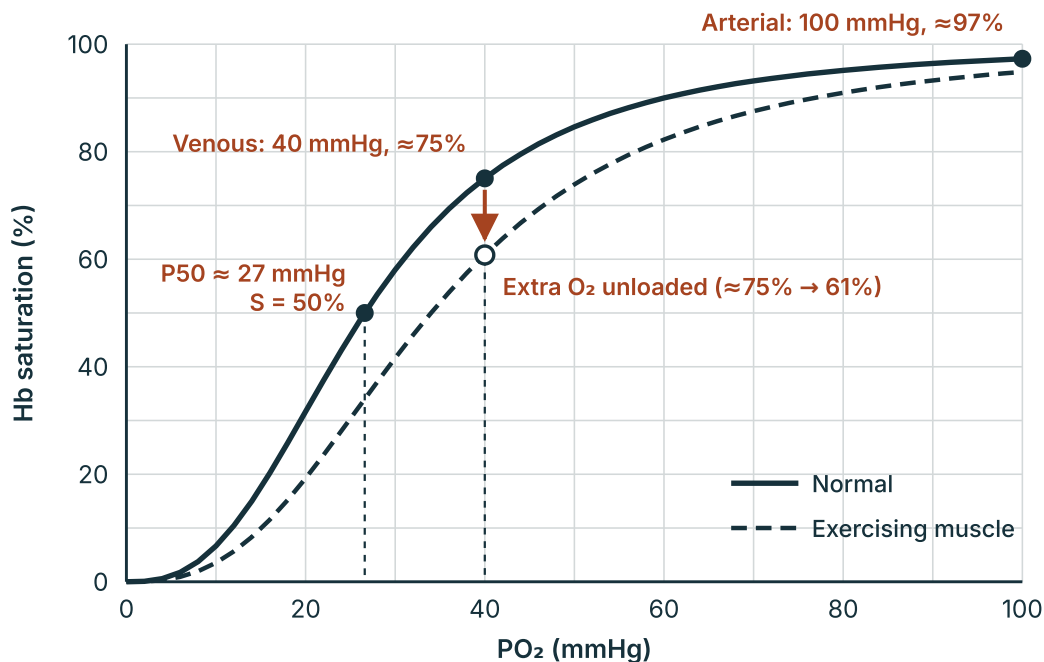
12 With CO ≈ 14.1 mL/dL; without CO ≈ 20.1 mL/dL; 30% lower. 1.34 × 15 × 0.70 = 14.07; 1.34 × 15 = 20.10; 14.07 / 20.10 = 0.70. (Real arterial blood is about 97% saturated, which scales both values by 0.97 and leaves the 30% drop unchanged.)

1. 1 pt: uses 70% of Hb as available for O₂.
2. 1 pt: 14.1 mL/dL (accept 14.0–14.1; 13.6 if 97% saturation was used).
3. 1 pt: 20.1 mL/dL without CO (19.5 at 97%) and a 30% drop.

O4 · transfer · S2, S3

Section D

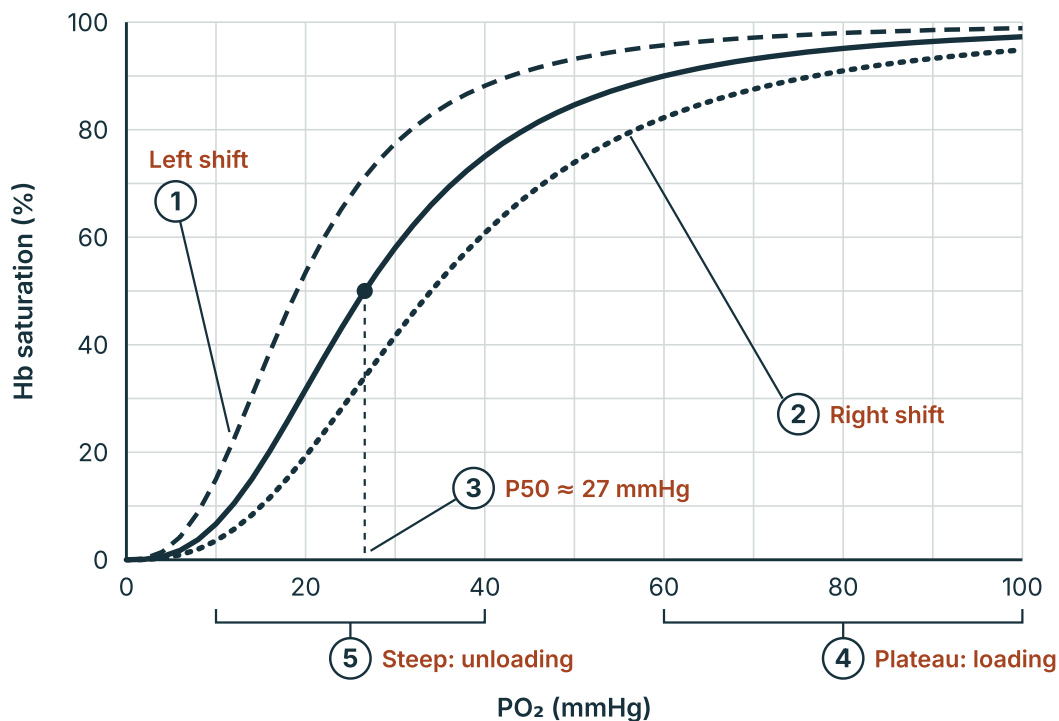
13 Model answer.



- 1 pt: sigmoid: flat near 0, steep from about 10 to 50 mmHg, plateau above about 70 mmHg.
- 1 pt: P₅₀ marked at about 27 mmHg, 50% (accept 25–28).
- 1 pt: arterial point at 100 mmHg, about 97%.
- 1 pt: venous point at 40 mmHg, about 75%.
- 1 pt: dashed curve to the right of the solid curve all along, reaching nearly the same plateau.
- 1 pt: arrow at 40 mmHg pointing down from solid to dashed curve (model: about 75% to 61%).

O1 (a), O2 (b) · S1, S2

14



1 Left-shifted curve (higher affinity, lower P50). Any one: fetal Hb, CO, low 2,3-BPG (stored blood), alkalosis or low PCO₂, low temperature. [1 — both the name and a condition are required]

2 Right-shifted curve (lower affinity, higher P50). Any one: high PCO₂, acidosis, high temperature, high 2,3-BPG (e.g., after days at altitude). [1 — both required]

3 P50 of normal blood, about 27 mmHg (accept 25–28). [1]

4 Plateau, the loading zone: large falls in PO₂ barely lower saturation, so loading in the lungs is protected. [1]

5 Steep part, the unloading zone: small falls in tissue PO₂ release large amounts of O₂. [1]

O1 · S1, S2 · Curves drawn from the Hill model with P50 19, 26.6 and 34 mmHg (schematic, $n = 2.7$).

AFTER THE TEST

Scoring sheet

Mark one column per item. **Unaided:** correct with no notes, hints or word bank. **With notes:** correct only after checking anything (circled items). **Missed:** wrong or blank. Partial credit goes in Earned; give an error type for anything short of full points.

ITEM	OBJECTIVE	PTS	UNAIDED	WITH NOTES	MISSED	EARNED	ERROR TYPE K · M · R · T
1	O1	2	☐	☐	☐	_____	_____
2	O1	2	☐	☐	☐	_____	_____
3	O2	2	☐	☐	☐	_____	_____
4	O5	2	☐	☐	☐	_____	_____
5	O3	2	☐	☐	☐	_____	_____
6	O2	2	☐	☐	☐	_____	_____
7	O3	2	☐	☐	☐	_____	_____
8	O3	2	☐	☐	☐	_____	_____
9	O2	3	☐	☐	☐	_____	_____
10	O3	3	☐	☐	☐	_____	_____
11	O4	4	☐	☐	☐	_____	_____
12	O4	3	☐	☐	☐	_____	_____
13	O1 · O2	6	☐	☐	☐	_____	_____
14	O1	5	☐	☐	☐	_____	_____

Total: _____ / 40 · **Unaided only:** _____ / 40 · **Time used:** _____ min

The unaided total is your exam estimate. Points earned with notes show exposure, not mastery.

AFTER THE TEST

Error analysis

Give every lost point one cause. The cause decides the fix; bring this page to your next study plan.

TYPE	WHAT IT LOOKS LIKE	FIX	ITEMS
K · Knowledge	Didn't know or misremembered the fact or concept.	Relearn it, then retrieve it: tutor drill, flashcards, retest in 1–3 days.	_____
M · Misread	Knew it; misread the stem, an option, units, or "NOT/EXCEPT."	Underline what the question asks and its qualifiers; answer before reading options.	_____
R · Reasoning	Knew the pieces; chained them wrong or picked a tempting distractor.	Worked examples and contrast items on that step; say why each wrong option is wrong.	_____
T · Time	Ran out of time or rushed it.	Timed mixed sets; set a per-item pace; skip and return.	_____

By objective

OBJECTIVE	POINTS LOST	MAIN TYPE	NEXT ACTION (AND WHEN)
O1 Read the curve	_____	_____	
O2 Shifts and the Bohr effect	_____	_____	
O3 Fetal Hb, stored blood, CO	_____	_____	
O4 Calculations	_____	_____	
O5 Diffusion direction	_____	_____	

Retest list

Items to redo cold in 2–3 days, without looking at the key:
